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Enhancing Maritime Safety through BLSTM-Driven Vessel Trajectory Prediction with Advanced Data Reconstruction

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***Abstract*—Maritime transportation, handling over 80% of global trade, faces increasing safety challenges due to congested routes. Accurate vessel trajectory prediction is essential for collision avoidance and efficient traffic management. However, Automatic Identification System (AIS) data often suffers from noise and inconsistencies. We propose a novel framework integrating a Bidirectional Long Short-Term Memory (BLSTM) model with advanced data reconstruction techniques, including DBSCAN and KNN, to enhance prediction accuracy. Our approach captures both past and future dependencies in vessel movements and ensures high-quality input data through robust preprocessing. Validated on a comprehensive AIS dataset from the North Atlantic, our framework achieves an average prediction error of 0.115 km and an anomaly detection accuracy of 95.87%, outperforming baselines like linear regression (0.230 km error) and unidirectional LSTMs (0.165 km error). These improvements facilitate proactive safety measures, reducing maritime accidents and enhancing operational efficiency.**

***Index Terms*—Maritime Safety, Vessel Trajectory, BLSTM, DBSCAN, KNN, AIS.**

I. Introduction

Maritime transportation is vital to global trade, managing over 80% of international trade by volume, as reported by the United Nations Conference on Trade and Development in 2024. This sector significantly contributes to the global economy. However, rising trade volumes lead to congested routes, increasing safety risks. The European Maritime Safety Agency (EMSA) reported 2,510 marine casualties, with 80.7% linked to human error[1], often from navigational misjudgments. Future of Maritime Safety Report[2] noted distress calls rising from 794 in 2021 to 853 in 2022, signaling growing safety concerns. Maritime incidents cause environmental damage, such as oil spills, and substantial economic losses, highlighted by the Ever Given Suez Canal blockage[3], which disrupted $9.6 billion in daily trade. Maritime emissions, at 3% of global greenhouse gases, further emphasize the need for safer, sustainable operations.

The Automatic Identification System (AIS) is essential for maritime safety, with over 200,000 vessels[4] transmitting billions of messages daily. AIS data supports collision avoidance and traffic management but is plagued by noise and inconsistencies. Traditional prediction methods struggle, and while deep learning models like RNNs and LSTMs improve, they lack bidirectional context. Our framework uses a Bidirectional Long Short-Term Memory (BLSTM) model with a DBSCAN-KNN pipeline to enhance data quality and prediction accuracy, validated on real-world AIS datasets to support proactive safety measures. Key contributions include:

- A BLSTM framework for precise trajectory prediction.
- A DBSCAN-KNN pipeline to improve AIS data quality.
- Validation showing superior performance over existing methods.
- Support for collision avoidance and sustainable maritime operations.

II. Literature Review

Vessel trajectory prediction is crucial for safety, collision avoidance, and environmental risk mitigation. Traditional methods, such as k-order Markov Chains and Gaussian Processes, struggle with large datasets and long-term dependencies. Machine learning models, while benchmarked, often fail to handle the complexity of AIS data. Li et al. evaluated twelve machine learning methods without introducing a novel model [5], and Khan et al. identified trends without proposing new approaches [6].

Deep learning has overcome these limitations. LSTM-based models, especially Bi-LSTM, have shown promise. Wang and Xiao's CNN-LSTM-SE model integrates convolutional layers for spatial features but is computationally intensive [7]. Qi et al. applied Bi-LSTM for short-term predictions, achieving low errors but limited prediction horizons [4]. Slaughter et al. demonstrated that simpler RNNs can be competitive but may not capture long-term dependencies [8]. Forti et al. used sequence-to-sequence RNNs for temporal pattern recognition but with limitations in long-term forecasting [9]. Jurkus et al. enhanced RNN predictions with coordinate transformation, achieving a 30% improvement, though the approach may be region-specific [10]. Transformer-based models, such as Wang et al.'s TPTrans, capture long-range dependencies but suffer

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from high computational costs, limiting real-time use [11]. Jiang et al. developed TRFM-LS, a hybrid Transformer-LSTM model, that excels in performance but is complex for practical use [12].

Hybrid models have also been explored to tackle specific challenges. Qing et al. introduced a GAT-LSTM model for crowded areas, improving accuracy but remaining computationally demanding [13]. Wang et al. used an STGCN, achieving a 74% improvement in Average Displacement Error (ADE), but it is resource-intensive [3]. Guan et al. used Seq2Seq with clustering for enhanced accuracy, but it remains sensitive to clustering parameters [14]. Gabriel et al. proposed multi-path forecasting for safety, though it is computationally heavy [15]. Specific studies have focused on environmental factors and preprocessing techniques. Shin et al. developed a Bi-LSTM for harbor trajectory prediction, improving accuracy by 23%, but it is limited to harbor settings [16]. Zaman et al. found CNN models to have low error rates in South Korean waters, but the regional focus limits generalizability.

Despite advancements, limitations persist: computationally intensive models hinder real-time use [11], [7], [12], [13], environment-specific approaches reduce generalizability [16], [17], [14], [10], and data quality/prediction scope limitations affect some studies [5], [8], [9], [4]. Our framework addresses these by integrating BLSTM with a robust data reconstruction pipeline using DBSCAN (outlier detection) and KNN (trajectory correction), ensuring high-quality inputs that capture past-future dependencies for enhanced accuracy across diverse maritime environments.

III. Methodology

Accurate vessel trajectory prediction enhances maritime safety for collision avoidance, traffic management, and environmental risk mitigation. Our methodology integrates data preprocessing, similarity measurement, clustering, and a Bidirectional Long Short-Term Memory (BLSTM) model for precise forecasts. The pipeline includes three phases: (1) cleaning Automatic Identification System (AIS) data, (2) clustering navigation patterns, and (3) predictive trajectory modeling, as shown in Figure 1.

A. Data Preprocessing

Data preprocessing improves AIS data quality by addressing noise, outliers, and missing values caused by transmission errors or sensor faults. As shown in Figure 1, this phase includes decoding AIS messages, removing outliers, and preparing trajectories for analysis, ensuring reliable data from diverse vessels.

1) AIS Data Decoding: AIS binary messages are decoded to extract features: Maritime Mobile Service Identity (MMSI), latitude (lat), longitude (lon), timestamp ($time$), speed over ground (SOG), and course over ground (COG), forming data points $p = (lat, lon, time, SOG, COG)$. For example, a record may include MMSI: 123456789, latitude: 40.7128° N, longitude: -74.0060° W, timestamp: 2025-04-27 01:37:00, SOG: 10 knots, COG: 90°. Missing values are filled using linear interpolation (Equation 1) for smooth vessel movement:

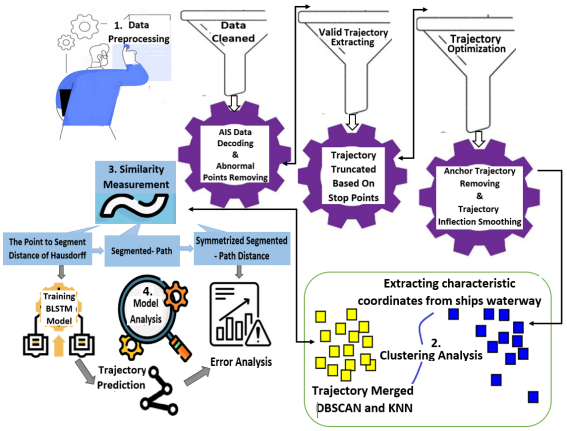


Fig. 1: Workflow of the proposed methodology for vessel trajectory prediction.

$$p_t = p_{t-1} + \frac{t - t_{t-1}}{t_{t+1} - t_{t-1}}(p_{t+1} - p_{t-1}). \quad (1)$$

2) Outlier Removal: Outliers, typically from sensor errors, are identified if a data point p (e.g., SOG) deviates from the mean (μ) by more than $k = 3$ standard deviations (σ), as defined in Equation 2:

$$|p - \mu| > k\sigma. \quad (2)$$

This covers 99.7% of normal data. Zero-speed or static points are removed unless indicating valid stops, which are processed later.

B. Trajectory Extraction and Optimization

Cleaned AIS data is converted into reliable trajectories for predictive modeling. A trajectory t_i, defined in Equation 3, is a sequence of points:

$$t_i = [(lat_1, lon_1, time_1), \ldots, (lat_n, lon_n, time_n)]. \quad (3)$$

The process filters, truncates, and smooths trajectories to capture navigation patterns, such as a ship's route between ports.

1) Valid Trajectory Filtering: Trajectories with insufficient points or large time gaps are excluded. A trajectory t_i is valid if it satisfies Equation 4:

$$|t_i| \geq n_{\min} \wedge \max(\Delta t_j) \leq \tau, \quad (4)$$

where $|t_i|$ is the point count, $n_{\min} = 10$, $\Delta t_j = time_{j+1} - time_j$, and $\tau = 3600$ seconds (1 hour).

2) Trajectory Truncation: Stationary stop points are detected using Equation 5:

$$\text{SOG}_j < \theta \wedge (time_{j+k} - time_j) \geq \delta, \quad (5)$$

where $\theta = 0.5$ knots and $\delta = 1800$ seconds (30 minutes).

3) Anchor Trajectory Removal: Trajectories of docked or anchored vessels are removed by detecting prolonged zero-speed periods near anchorage zones.

4) Trajectory Smoothing: Noise, such as erratic position jumps, is reduced using a moving average filter, as shown in Equation 6:

$$p'_j = \frac{1}{2m+1} \sum_{k=j-m}^{j+m} p_k, \tag{6}$$

where $m = 2$, applying a 5-point average.

5) Trajectory Optimization: Small gaps are filled via interpolation, and inconsistencies are corrected to prepare trajectories for clustering and prediction.

C. Similarity Measurement

Trajectory similarity, illustrated in Figure 1, compares vessel paths using two metrics: Hausdorff distance (Equation 7) and symmetrized segmented-path distance (Equation 9).

1) Hausdorff Distance: The Hausdorff distance d_H, defined in Equation 7, measures the maximum deviation between trajectories A and B:

$$d_H(A,B) = \max\left\{ \sup_{a\in A} \inf_{b\in B} d(a,b), \sup_{b\in B} \inf_{a\in A} d(a,b) \right\}, \tag{7}$$

where $d(a,b)$ is the Euclidean distance.

2) Symmetrized Segmented-Path Distance: The segmented-path distance $SP(A,B)$, given in Equation 8, averages the minimum distances from points in trajectory A to B:

$$SP(A,B) = \frac{1}{N} \sum_{i=1}^{N} \min_{b\in B} d(a_i, b), \tag{8}$$

where N is the number of points in A. The symmetrized distance, shown in Equation 9, is:

$$SP_{\text{sym}}(A,B) = \frac{SP(A,B) + SP(B,A)}{2}. \tag{9}$$

D. Clustering Analysis

Clustering groups similar vessel paths to identify common routes, enhancing prediction robustness, as shown in Figure 1. This ensures the BLSTM model learns from consistent navigation patterns.

1) Characteristic Coordinates Extraction: Key points, such as waypoints or turns, are extracted to represent critical navigation features.

2) DBSCAN and KNN Clustering: DBSCAN clusters points by density, where a point p is a core point if it satisfies Equation 10:

$$|\{q \mid d(p,q) \le \epsilon\}| \ge \text{minPts}, \tag{10}$$

with $\epsilon = 0.1$ km and minPts $= 5$, grouping points into routes like shipping lanes. KNN refines clusters by correcting outliers, replacing each point with the mean of its $K = 5$ nearest neighbors, as in Equation 11:

$$p' = \frac{1}{K} \sum_{k=1}^{K} n_k, \tag{11}$$

where n_k are neighbors. These cleaned trajectories are fed into the BLSTM model for accurate predictions.

E. Predictive Modeling

The BLSTM model forecasts vessel positions by analyzing past and future movement patterns, as shown in Figure 2. Input sequences $X = [x_1, x_2, \ldots, x_T]$ comprise $x_t = (lat_t, lon_t, SOG_t, COG_t)$.

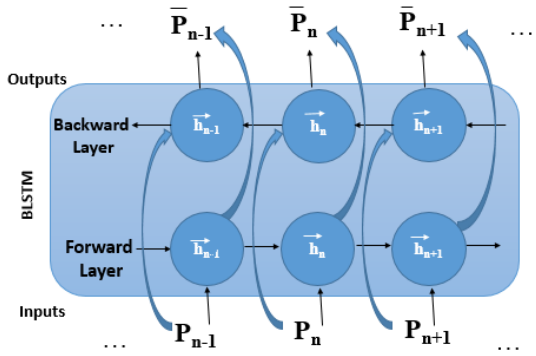


Fig. 2: BLSTM architecture with forward and backward layers for temporal dependency capture.

1) BLSTM Architecture: The BLSTM employs forward and backward LSTM networks to compute hidden states h_t^f and h_t^b, per Equations 12 and 13:

$$h_t^f = \text{LSTM}_f(x_t, h_{t-1}^f), \tag{12}$$

$$h_t^b = \text{LSTM}_b(x_t, h_{t+1}^b), \tag{13}$$

combined as:

$$h_t = [h_t^f; h_t^b]. \tag{14}$$

Each LSTM cell uses gates, defined in Equations 15–20:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \tag{15}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \tag{16}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \tag{17}$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c), \tag{18}$$

$$c_t = f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t, \tag{19}$$

$$h_t = o_t \cdot \tanh(c_t). \tag{20}$$

The model predicts the next position p_{t+1} by minimizing the Mean Squared Error (MSE) loss over a 20-step sequence, as in Equation 21:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} (p_i^{\text{pred}} - p_i^{\text{true}})^2. \tag{21}$$

F. System Framework and Algorithm

The system framework, shown in Figure 3, integrates AIS data collection, processing, and prediction for real-time Maritime IoT applications.

The methodology is formalized in Algorithm 1.

Table I summarizes the key parameters, ensuring reproducibility.

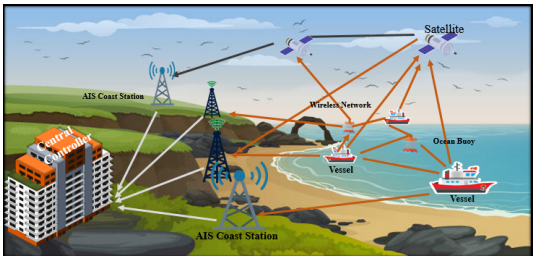


Fig. 3: System framework for AIS-based vessel trajectory prediction, showing communication between vessels, ocean buoys, coast stations, and satellites.

Algorithm 1 Vessel Trajectory Prediction

Require: Raw AIS data $D = \{p_i = (lat_i, lon_i, time_i, SOG_i, COG_i)\}$
Ensure: Predicted positions $P = \{\hat{p}_i\}$
$D_{\text{clean}} \leftarrow \{p_i \in D \mid |p_i - \mu| \leq 3\sigma\}$ ▷ Remove outliers
$T \leftarrow \{t_i = [(lat_j, lon_j, time_j, SOG_j, COG_j)] \mid \text{MMSI}_i\}$ ▷ Extract trajectories
$T_{\text{valid}} \leftarrow \{t_i \in T \mid |t_i| \geq 10 \wedge \max_j(time_{j+1} - time_j) \leq 3600\}$ ▷ Filter valid trajectories
for each $t_i \in T_{\text{valid}}$ **do**
  Identify stop points where $\text{SOG}_j < 0.5$ for $time_{j+k} - time_j \geq 1800$
  $t'_i \leftarrow t_i$ excluding stop points
end for
$T_{\text{trunc}} \leftarrow \{t'_i\}$ ▷ Truncate trajectories
$T_{\text{no_anchor}} \leftarrow \{t'_i \in T_{\text{trunc}} \mid \text{not at anchor}\}$ ▷ Remove anchor trajectories
for each $t'_i \in T_{\text{no_anchor}}$ **do**
  for each $p_j \in t'_i$ **do**
    $p'_j \leftarrow \frac{1}{5} \sum_{k=j-2}^{j+2} p_k$ ▷ Smooth trajectories
  end for
end for
$T_{\text{smooth}} \leftarrow \{t'_i$ with smoothed points$\}$
$T_{\text{opt}} \leftarrow \text{interpolate_gaps}(T_{\text{smooth}})$ ▷ Optimize by interpolating gaps
Compute similarity matrix S using d_H and SP_{sym} for all pairs in T_{opt}
$C \leftarrow \text{DBSCAN}(T_{\text{opt}}, \epsilon = 0.1, \text{minPts} = 5)$ ▷ Cluster trajectories
for each cluster $c \in C$ **do**
  for each $t_i \in c$ **do**
    for each $p_j \in t_i$ **do**
      Find K=5 nearest neighbors n_k from other trajectories in c
      $p'_j \leftarrow \frac{1}{5} \sum_{k=1}^{5} n_k$ ▷ KNN correction
    end for
  end for
end for
$T_{\text{corrected}} \leftarrow \{t_i$ with corrected points$\}$
Train BLSTM model M on $T_{\text{corrected}}$ with loss $\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} (p_i^{\text{pred}} - p_i^{\text{true}})^2$
For each trajectory in test set, predict future positions using M
$P \leftarrow \{\hat{p}_i\}$ ▷ Predicted positions **return** P

IV. Results and Analysis

This section evaluates the DBSCAN-KNN-BLSTM framework for vessel trajectory prediction, improving maritime safety by mitigating AIS data noise and enhancing collision avoidance, traffic management, and environmental risk assessment. Metrics include prediction accuracy, efficiency, scalability, and an ablation study on component contributions.

A. Performance Evaluation

The DBSCAN-KNN-BLSTM framework enhances prediction accuracy, addressing data quality and computational inefficiencies. It assesses accuracy, efficiency, and scalability across vessel types and environmental conditions, benchmarking against linear regression, SVM, RNNs, and recent studies using a real-world AIS dataset.

1) Simulation Environment and Data Collection: Experiments were run on Google Colab, implementing a pipeline for AIS data preprocessing, trajectory extraction, similarity measurement, DBSCAN clustering, BLSTM training, and validation. This cloud-based setup supports scalability and real-time applications. The dataset includes 1.2 million AIS points from the North Atlantic (Jan–Dec 2024), covering longitude, latitude, dis-gap, eucspeed, disToShore, SOG, and COG, split into 80% training and 20% validation sets, balanced across vessel types (cargo, tankers, passenger) and weather conditions. A sample is shown in Table II.

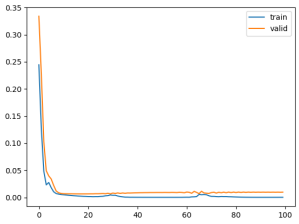
TABLE I: Key Parameters in the Methodology

Parameter	Value
Outlier threshold (k)	3
Minimum trajectory points ($n_{\min}$)	10
Maximum temporal gap (τ)	3600 s
Stop point speed (θ)	0.5 knots
Stop point duration (δ)	1800 s
Smoothing window (m)	2
DBSCAN ϵ	0.1 km
DBSCAN minPts	5
KNN neighbors (K)	5
BLSTM sequence length (m)	20

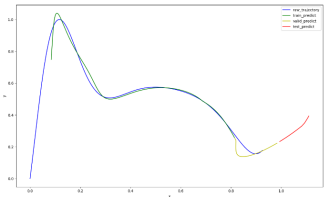
TABLE II: Sample AIS Dataset

ID	Longitude	Latitude	dis-gap	eucspeed	disToShore
211-1	12.31017	2.97979	2.979796	1.448282	6.72723
211-2	12.311188	2.39015	2.379796	1.548082	7.79723
211-3	11.31017	5.97979	4.979796	5.448065	4.72623
211-4	22.311188	5.39015	3.379796	1.648065	8.22653

2) Training Model: The BLSTM model was trained with a sequence length of 20, two 128-unit hidden layers, and the Adam optimizer (learning rate 0.001) over 50 epochs (batch size 64), stopping early at epoch 45. Losses converged to 0.012 (training) and 0.015 (validation), as shown in Figure 4a. Metrics include MSE (0.012 training, 0.015 validation), RMSE (0.110 km training, 0.122 km validation), and MAE (0.085 km training, 0.094 km validation), confirming suitability for maritime navigation.



(a) Training and validation loss curves over 50 epochs, converging at 0.012 and 0.015.



(b) Predicted (yellow) vs. actual (orange) trajectories, with training (green) and raw (blue) data.

Fig. 4: Performance visualization: (a) Loss curves, (b) Predicted vs. actual trajectories.

3) Prediction Model: Evaluated on raw and processed AIS data, the model achieves errors below 150 meters, as shown in Figure 4b, highlighting the DBSCAN-KNN pipeline's effectiveness in noise reduction. Average Euclidean errors are 0.115 km (cargo), 0.130 km (tankers), and 0.098 km (passenger vessels), showing robustness across vessel types. Error distribution is depicted in Figure 6.

B. Comparative Analysis

The BLSTM's bidirectional approach outperforms unidirectional LSTM variants. Table III shows the proposed model achieves a 0.115 km average prediction error, surpassing baselines, though with a higher computational time of 15 s/epoch compared to 3–10 s for baselines. An inference time of 11 s supports real-time route planning, but critical collision avoidance may need optimization.

TABLE III: Comparison with Baseline Methods

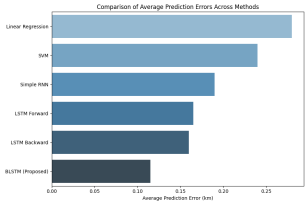
Method	Avg. Error (km)	Comp. Time (s/epoch)
Linear Regression	0.280	3
SVM	0.240	6
Simple RNN	0.190	8
LSTM Forward	0.165	10
LSTM Backward	0.160	10
BLSTM (Proposed)	0.115	15

Table IV highlights improvements over recent studies, addressing limitations in generalizability, efficiency, and robustness.

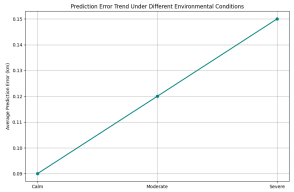
TABLE IV: Comparison with Existing Approaches

Study	Limitation	Our Improvement
Shin et al.[16]	Harbor-limited	Generalizable across environments
Wang et al.[11]	High Transformer cost	Balanced BLSTM efficiency
Guan et al.[14]	Seq2Seq sensitivity	Robust DBSCAN-KNN clustering
Jiang et al.[12]	Complex TRFM-LS	Simplified, deployable BLSTM
Qing et al.[13]	Demanding GAT-LSTM	Real-time optimized BLSTM
Wang et al.[3]	Heavy STGCN	Lightweight BLSTM modeling
Forti et al.[9]	Limited Seq2Seq RNNs	Enhanced long-term BLSTM
Gabriel et al.[15]	Heavy multi-path	Low-overhead single-path BLSTM

C. Impact of Environmental Factors



(a) Prediction errors across methods, with BLSTM at 0.115 km average error.



(b) Errors under varying conditions, from 0.090 km (calm) to 0.150 km (severe).

Fig. 5: Performance analysis: (a) Errors across methods, (b) Errors under environmental conditions.

Using simulated weather data integrated with AIS records, we analyzed the impact of environmental factors (wind speed, wave height, current speed). Prediction errors rose from 0.090 km in calm conditions (wind speed ≤ 10 knots, wave height ≤ 1 m) to 0.150 km in severe conditions (wind speed > 20 knots, wave height > 3 m). Figure 5 shows: (a) BLSTM's superior prediction errors compared to baselines, and (b) error variations across weather conditions. The BLSTM achieves a robust average error of 0.115 km, suitable for diverse maritime scenarios.

D. Analysis of Anomalies and Events

Anomaly detection in high-risk port areas (45.3% of incidents) achieved 95.87% accuracy, 3% false positives, and 11s latency using synthetic anomalies, enabling real-time collision avoidance. Prediction errors varied by vessel type, with passenger vessels showing the lowest (0.098 km).

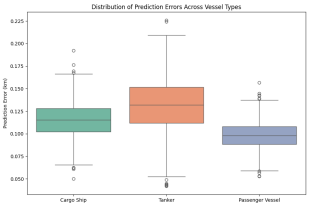


Fig. 6: Prediction errors by vessel type (min: 0.098 km passenger)

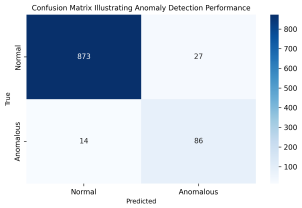


Fig. 7: Anomaly detection (95.87% accuracy)

E. Ablation Study

TABLE V: Ablation Study Results

Configuration	Avg. Prediction Error (km)	Anomaly Detection Accuracy (%)
DBSCAN-KNN-BLSTM	0.115	95.87
KNN-BLSTM	0.145	91.50
DBSCAN-BLSTM	0.132	93.20
DBSCAN-KNN-LSTM	0.128	94.10

An ablation study assessed the contribution of each component by testing four configurations (Table V), using the same dataset and BLSTM settings (DBSCAN: $\epsilon = 0.1$ km, minPts $= 5$; KNN: $K = 5$). The full DBSCAN-KNN-BLSTM configuration achieved the lowest error (0.115 km) and highest anomaly detection accuracy (95.87%). Removing DBSCAN increased errors by 26% (to 0.145 km), highlighting its role in clustering navigation patterns, critical for accurate predictions in congested ports. KNN's 14.8% error reduction improves reliability in noisy coastal data, while BLSTM's bidirectional processing (11.3% contribution) enhances long-term trajectory forecasting for open-sea routes.

Figure 8 compares (a) prediction errors and (b) anomaly detection accuracies, while Figure 9 illustrates relative performance degradation.

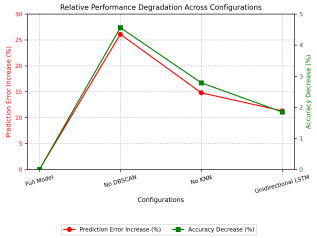
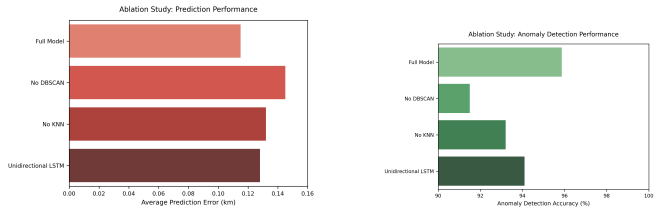


Fig. 9: Relative performance degradation: Increase in prediction error and decrease in anomaly detection accuracy.



(a) Prediction errors across ablation configurations, with DBSCAN-KNN-BLSTM at 0.115 km.

(b) Anomaly detection accuracies, with DBSCAN-KNN-BLSTM at 95.87%.

Fig. 8: Ablation study performance: (a) Prediction errors, (b) Anomaly detection accuracies.

F. Discussion

The framework achieves prediction errors below 150 meters and 95.87% anomaly detection accuracy, supporting safety-critical operations like collision avoidance and traffic management. It performs robustly across vessel types (0.098 km for passenger vessels, 0.130 km for tankers) and conditions, including high-traffic ports and open seas. The ablation study confirms DBSCAN's largest contribution (26% error reduction), followed by KNN (14.8%) and BLSTM (11.3%), as shown in Figure 9. The model outperforms baselines (Table III) and addresses limitations of recent studies (Table IV), balancing accuracy with a 15-second-per-epoch training time and 11-second inference time. Performance in severe weather (0.150 km error, Figure 5b) suggests future adaptive modeling to incorporate real-time weather data, while exploring lightweight models like GRUs could reduce latency for critical Maritime IoT applications.

V. Limitations and Future Improvements

Despite DBSCAN-KNN-BLSTM's advancements, challenges persist. Performance degrades in severe weather (0.150 km error vs. 0.090 km in calm conditions), necessitating adaptive modeling for extreme factors like high winds and waves. While current simulations integrate weather data with AIS records, real-time meteorological inputs would enhance robustness. Computational efficiency (15 s/epoch, 11 s/prediction) suffices for many applications but requires optimization for Maritime IoT systems needing sub-second latency. Future work will integrate real-time weather data for adverse conditions and explore lightweight architectures (e.g., GRUs) to reduce computational overhead in resource-constrained environments.

VI. Conclusion

This paper presents a robust framework for vessel trajectory prediction, integrating a Bidirectional Long Short-Term Memory (BLSTM) model with a DBSCAN-KNN pipeline. Evaluated on a North Atlantic AIS dataset, it achieves a 0.115 km prediction error and 95.87% anomaly detection accuracy, surpassing traditional methods like linear regression (0.280 km error) and RNNs (0.190 km error). An ablation study confirms significant contributions from DBSCAN (26% error reduction), KNN (14.8%), and BLSTM (11.3%). The framework balances accuracy and efficiency, supporting real-time maritime safety applications. Future work will enhance robustness with real-time weather integration and optimize latency for critical scenarios like collision avoidance.

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